COMPETITION BETWEEN FOSSIL AND RENEWABLE ENERGY IN ARDEN

Arden State Energy Office, occasional paper

1. THE QUESTION AND THE ANSWER

The Office is asked whether a charge on fossil fuel sales advantages Arden's renewable sector. The short answer is that the two compete in some end uses and not in others, and that the Office cannot quantify the effect of a price change in one on volumes in the other.

2. WHERE THEY COMPETE

In electricity generation they compete directly, and the Office estimates the elasticity of substitution as material at the margin. In space and water heating they compete for new installations but not for the existing stock, of which 71.6 per cent is gas-fired with an average remaining life of 14 years. In transport they do not presently compete at scale; the electric share of new registrations is 10.3 per cent and of the fleet 3.4 per cent.

3. WHAT THE ARDEN RENEWABLE SECTOR SELLS

A majority of the sector's revenue is equipment manufacture and services, much of it sold outside Arden. Generation revenue is the smaller part. A change in the in-state price of fossil fuel would affect the generation part directly and the equipment part only through national demand, in which Arden's share is 2.7 per cent.

4. PUBLIC SUPPORT

The sector receives generation credits, manufacturing grants and residential rebates, all from the general fund. The Office notes that none of these is funded from charges on competing products and that the Resilience Fund could not be applied to them, being confined to repair and adaptation projects.

5. WHAT THE OFFICE WOULD NEED TO MODEL THE BILL

Three things the Office does not have. The pass-through rate from a charge on the seller to the retail price, which depends on the elasticity of demand and on the competitive structure of each product market. The cross-price elasticity between fuel and the electricity that would substitute for it, which differs by end use and by season. And the investment response of generators, which is a matter of expectation rather than of observed behaviour. Absent all three, any figure the Office produced would be an assumption dressed as an estimate.

6. WHAT CAN BE SAID

Two things, both weak. The charge on annual revenues is small relative to the price variation this market absorbs routinely: the volume-weighted price of refined products in Arden varied by about 37.2 per cent peak to trough over the measuring period, which is orders of magnitude larger than half of one per cent. And the one-time assessment is not a marginal charge at all — it attaches to sales already made and cannot alter the price of those sales, whatever it does to the payer's balance sheet.

It follows that if the bill affects the relative position of renewable generation it does so through the annual charge and not through the assessment, and the annual charge is small. The Office does not draw the further conclusion that the effect is nil, only that it has no basis for quantifying it.

7. LIMITS OF THIS PAPER

The Office has not modelled the effect of the bill and does not propose to. Any estimate would require the three assumptions set out in section 5, none of which the Office can supply from its own data.

Occasional papers represent the analysis of staff and not the position of the Office.